

How Sky Score works, in plain English

What happens when you type a postcode, and what the Amazon pieces do · 2026-09-21

In one breath: the website is a page with a big table of numbers built in. When you type a postcode, it asks a free public service *where* that postcode is, looks up the borough in its own table, does some sums, and shows you a score. Then it asks four small helpers for extras — stations, sold prices, energy ratings, GPs. No artificial intelligence is involved on the website at all.

Part 1 — What happens when you type N1 7SX

Think of your browser as a person at a desk with a reference book. Here is the conversation it has, in the order it happens. The whole thing takes about a second.

YOUR BROWSER, TO ITSELF

"Someone's typing. I'll fetch the three lists I'll need in a moment: the aircraft-noise readings for London, the same for the regions, and the list of stations."

Three files come down from the website's own shelf. This happens once per visit.

YOUR BROWSER → POSTCODES.IO (A FREE PUBLIC SERVICE)

"Where is N1 7SX?"

Answer: "Latitude 51.53, longitude -0.09. It is in **Hackney**." postcodes.io is a free lookup built on the government's postcode register. It knows where every UK postcode is.

YOUR BROWSER, TO ITSELF

"Hackney. My table has a row for Hackney: house prices, crime, schools, transport, air quality, road noise, flood risk. I'll do the sums."

The score is worked out **right here, on your laptop**. The table of numbers came with the page when it loaded. Nothing is sent anywhere for this step.

YOUR BROWSER, TO ITSELF

"For aircraft noise I can do better than the borough. Is N1 7SX in the noise-readings list?"

If the government measured aircraft noise at this exact postcode, that reading is used. If not, the browser estimates it from how far the postcode is from the flight paths. This is the only number that is about your exact postcode; everything else is about the borough, and the page says so.

Now it knows the location, so it sends four questions at the same time, to four small helpers I run on Amazon:

YOUR BROWSER → HELPER 1 ("TRANSPORT")

"Which stations are near this spot, and are the lines running?"

"Angel, 780 m. Essex Road, 848 m. Old Street, 916 m. Northern line: good service."

YOUR BROWSER → HELPER 2 ("NHS")

"Any GPs, pharmacies or clinics within a 20-minute walk?"

"Prebend Street Surgery, 407 m. St Peter's Street Medical Practice ..."

YOUR BROWSER → HELPER 3 ("SOLD-PRICES")

"What has sold at this postcode?"

"Windsor House, Wenlock Road: £490,000 in 2015; £190,000 in 2011 ..." (from HM Land Registry)

YOUR BROWSER → HELPER 4 ("EPC")

"Any energy certificates at this postcode?"

"79 of them. Average band C." (from the government's energy-certificate register)

YOUR BROWSER, TO ITSELF

"Got everything. Draw the panel."

Score and the four component scores at the top (from its own sums), then stations, sales, energy, NHS (from the four helpers).

your browser, on its own

a free public service

a small helper I run on Amazon

The important thing it did not do: it never asked anyone for the score. The score is sums done on your laptop, over numbers that were worked out weeks ago from government files. Businesses who use the API get the same sums from Amazon instead — and before any change is saved, the website and the API are checked against each other for every borough to make sure they still agree.

What ends up on the screen for N1 7SX

5.4 / 10 — Mixed

"Mixed picture. This location scores well on some factors but is held back by high prices."

Quiet Skies (aircraft noise)	7.0 / 10
Affordability	1.1 / 10
Growth (prices, adjusted for inflation)	3.8 / 10
Liveability (crime, schools, transport, healthcare)	7.8 / 10
Environment (air, road noise, flood)	5.2 / 10

Under the scores it says: "Quiet Skies uses your location's noise data. Other scores are Hackney borough averages." That sentence is the honesty line: it tells you which number is about your postcode (one) and which are about your borough (the rest).

Part 2 — The Amazon pieces, and why each one is there

Sky Score runs on Amazon Web Services (AWS) — Amazon's business of renting out computers and storage by the minute. We rent nine kinds of thing. None of them is a computer that sits switched on all day. Each is described below the same way: **what it is**, in everyday terms; **what we use it for**; and **why** this and not something else.

S3 the filing cabinet

WHAT IT IS

A place to keep files. You put a file in; it stays there until you take it out.

WHAT WE USE IT FOR

Every file the website is made of: the page itself, the map outlines, the table of borough numbers, the noise readings, the station list, and 100 small pages, one per borough.

WHY

A website that is *just files* is the cheapest and hardest-to-break kind there is. There is no program running to crash. Storing all of it costs pennies a month.

CloudFront photocopies in every city

WHAT IT IS

A network of Amazon buildings around the world that each keep a copy of your files. A visitor gets the copy nearest to them.

WHAT WE USE IT FOR

Serving skyscore.co.uk. Someone in Manchester gets the page from a copy in the UK, not from the filing cabinet in London every time. It also keeps a copy of the little score badge that other websites can embed.

WHY

Speed and cost: copies close to the visitor arrive faster, and the filing cabinet is bothered less. It also lets us add the security settings every serious website needs, in one place.

API Gateway the receptionist

WHAT IT IS

The front desk for the business side of the service. Every request from a business, or from the website's four helpers, comes through here first.

WHAT WE USE IT FOR

Checking the caller's pass (the "API key"), counting how many times they have called this month against the free allowance (10,000), slowing anyone who hammers the door, and sending each request to the right worker.

WHY

Because these jobs are boring, important and easy to get wrong. A receptionist that Amazon runs and that never sleeps is better than one I write myself.

Lambda workers on call — eight of them

WHAT IT IS

A small program that does not run all the time. It is *woken up* when there is a job, does the job in a fraction of a second, and goes away again. You pay per job, not per day.

WHAT WE USE IT FOR

Eight jobs, one worker each: **score** (the business API), **chat** (the assistant), **signup** (handing out passes), **favourites** (saved places), and the four helpers the website uses: **transport**, **nhs**, **sold-prices**, **epc**.

WHY

A normal server is like renting an office 24 hours a day even when nobody is in it. Workers on call cost nothing when nobody is asking, and there is no office to clean, patch or restart. For a service that is quiet most of the time, this is the cheapest way there is.

DynamoDB the index-card box

WHAT IT IS

A very fast box of index cards. Give it a card's number and it hands you the card at once, however many cards there are.

WHAT WE USE IT FOR

Five boxes. The big one holds 2.7 million cards, one per UK postcode, each saying where it is and which borough it is in. Another holds the government's noise readings, one per postcode. The other three are small: saved places, sign-ups, and sign-ups waiting to be confirmed.

WHY

We only ever ask one question of it: "the card for this postcode, please." We never need to search across cards or add them up. So we do not need a full database with a program running all day — just a fast box, paid for by the lookup.

Bedrock the door to the AI model

WHAT IT IS

Amazon's service for talking to AI language models without running one yourself. You send text in, text comes back.

WHAT WE USE IT FOR

One thing only: the assistant on the business API. It is handed the score for a postcode and asked to explain it in a sentence. The model is Amazon Nova 2 Lite, a small, cheap one.

WHY

The job is reading numbers back in English, which does not need a clever model — a cleverer one invents more fluently, not less. This is also the *only* piece of Sky Score outside the UK: the model lives in Amazon's Virginia region, so the question and the score numbers travel there. Nothing personal does, and the privacy page says so.

CloudWatch the logbook

WHAT IT IS

Every worker writes down what it did. CloudWatch is where those notes go, kept for 30 days.

WHAT WE USE IT FOR

Finding out what happened when something goes wrong — and seeing the assistant's check catch the model inventing a number. Each catch is one line in the logbook, with the numbers that were refused.

WHY

Workers on call leave nothing behind when they finish, so without a logbook there would be no way to look back. Thirty days is enough to investigate and short enough to keep the privacy promise.

IAM the keys and passes

WHAT IT IS

Who is allowed to do what. Every worker, and every person, has a written list of what they may touch.

WHAT WE USE IT FOR

One "deploy" identity, with one written list, is allowed to put new versions of the workers and files in place. Each worker is allowed only what its job needs — the chat worker may knock on Bedrock's door; nobody else may.

WHY

If something is broken into or goes wrong, the damage is limited to what that one identity was allowed to do. It also means the coding assistant I build with cannot change the live system by accident — putting a new version live needs my explicit go-ahead, and on one occasion its own safety check refused and I ran the step myself.

CloudFormation, via "SAM" the blueprint

WHAT IT IS

A single document that describes every piece above — each worker, each card box, the receptionist's rules. "Deploying" means telling Amazon: make the real thing match this document.

WHAT WE USE IT FOR

Everything on the business side is set up from that one file. Nothing was clicked together by hand in a console and then forgotten.

WHY

Because a thing set up by hand cannot be rebuilt, reviewed or checked. A blueprint can be read, tested, and put back exactly if something is lost.

Two things we deliberately do not rent

1

No always-on computer. There is no server to log into. Every piece above either stores files or wakes up for a job.

- 2 **No map service.** The map is drawn from our own outlines, in the browser. No Google Maps, no map key, no charge per view, and the site still works with no internet once it has loaded once.

Why Amazon, in three sentences

The project began at an Amazon-run hackathon, so Amazon's tools were the starting point, and the AI model it uses only exists there. Everything above can be paid for by the job rather than by the day, which suits a service that is quiet most of the time. And one account, one bill and one blueprint is simpler to keep honest than a mixture of providers.

What got taken out, and why

In March there were five AI features on the website: a chat panel, an automatic AI summary on every postcode, photo analysis, document analysis and a generated report. In May all five were removed in one go. The main reason: an AI summary on top of a carefully checked score adds an answer that varies and is sometimes wrong, and for a product about someone's home a wrong number costs more than a dull one. The second reason: the five were costing about \$80–115 a month in credits. What came back in August is one narrow assistant that can only explain numbers it was handed, for about 25p per thousand questions.

Part 3 — How to show Part 1 live in 30 seconds

- 1 Open skyscore.co.uk in Chrome. Press **F12**. A panel opens: click **Network** at the top of it, then the **Fetch/XHR** button so only data requests are shown. Click the small 🗑 icon to empty the list.
- 2 Type **N1 7SX** in the search box and press Enter.
- 3 Watch the list fill: three files, postcodes.io, then the four helpers landing together. Read them out. Then: "and notice there is no request for the score."
- 4 Tip: drag the panel to the right-hand side (the three dots in its corner → Dock side) so the map and the list are visible together.

Appendix — the same conversation as the browser writes it

These nine lines are what the Network panel shows. Captured live on 2026-09-21 while searching SW11 1AA; the same nine appear for any UK postcode.

#	WHAT THE BROWSER ASKED FOR	PLAIN ENGLISH
1	/data/aircraft-quiet-london.json	London aircraft-noise readings (35,352 postcodes)
2	/data/aircraft-quiet-regions.json	the same for seven regional airports (7,339 postcodes)
3	/data/stations.json	the station list (1,390 stops, ten cities)
4	api.postcodes.io/postcodes/SW111AA/autocomplete	the suggestions as you type
5	api.postcodes.io/postcodes/SW111AA	"where is this postcode?"
6	/transport?lat=51.464&lon=-0.164	helper 1: stations and line status
7	/nhs?lat=51.464&lon=-0.164	helper 2: GPs and clinics nearby
8	/sold-prices?postcode=SW11 1AA	helper 3: recent sales
9	/epc?postcode=SW11 1AA	helper 4: energy certificates

Words used above

Browser

Chrome, Safari, Edge — the program you view websites in. It does more than display pages; it can do sums and ask questions of other computers.

Request

One question the browser sends over the internet, and the one answer that comes back.

API

A way for one program to ask another for data, without a person or a screen in between. Sky Score's API: send a postcode, get numbers back.

API key

A pass that identifies who is calling, so their usage can be counted.

Deploy

Put a new version live.

Region

One of Amazon's clusters of buildings. Sky Score uses London; the AI model is only in Virginia.

Sky Score · requests captured live and panel values for N1 7SX read from the live site, both 2026-09-21. The removal figures are from the commit of 7 May 2026 that made the change.